SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
Constituent of Symbiosis International (Deemed University), Pune
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वसुधैव कुटुम्बकम् Re-Accredited by NAAC with 'A++' Grade

ANISH NAIR

BATCH: A1

PRN:25070122037

Subject: Programming with Java

ASSIGNMENT No: 7

TITLE : Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples**.**

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction**.**

Theory

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Outcome

```java
abstract class Shape {
    String name;

    Shape(String name) {
        this.name = name;
    }

    abstract double calculateArea();

    void display() {
        System.out.println(name + " Area: " + calculateArea())
    }
}

class Circle extends Shape {
    double radius;

    Circle(double radius) {
        super("Circle");
        this.radius = radius;
    }

    @Override
    double calculateArea() {
        return Math.PI * radius * radius;
    }
}

class Rectangle extends Shape {
    double length, breadth;

    Rectangle(double length, double breadth) {
        super("Rectangle");
        this.length = length;
        this.breadth = breadth;
    }

    @Override
    double calculateArea() {
        return length * breadth;
    }
}
```

```java

class Circle extends Shape {
    double radius;

    Circle(double radius) {
        super("Circle");
        this.radius = radius;
    }

    @Override
    double calculateArea() {
        return Math.PI * radius * radius;
    }
}

class Rectangle extends Shape {
    double length, breadth;

    Rectangle(double length, double breadth) {
        super("Rectangle");
        this.length = length;
        this.breadth = breadth;
    }

    @Override
    double calculateArea() {
        return length * breadth;
    }
}

public class ShapeDemo {
    public static void main(String[] args) {
        Shape circle = new Circle(5.0);
        Shape rectangle = new Rectangle(4.0, 6.0);

        circle.display();
        rectangle.display();
    }
```

```
Circle Area: 78.53981633974483
Rectangle Area: 24.0
```

```java
// ==================== File: ECommerceProductDemo.java ====================
interface Product {
    void showDetails();
    double getPrice();
}

class BaseProduct {
    String name;
    double price;

    BaseProduct(String name, double price) {
        this.name = name;
        this.price = price;
    }
}

class ElectronicProduct extends BaseProduct implements Product {
    int warrantyYears;

    ElectronicProduct(String name, double price, int warrantyYears) {
        super(name, price);
        this.warrantyYears = warrantyYears;
    }

    @Override
    public void showDetails() {
        System.out.println("Electronic Product: " + name + " | Price: Rs. " + price + " | Warr
    }

    @Override
    public double getPrice() {
        return price;
    }
}

class ClothingProduct extends BaseProduct implements Product {
    String size;

    ClothingProduct(String name, double price, String size) {
        super(name, price);
        this.size = size;
```

```java
class ClothingProduct extends BaseProduct implements Product {
    ClothingProduct(String name, double price, String size) {
    }

    @Override
    public void showDetails() {
        System.out.println("Clothing Product: " + name + " | Price: Rs. " + price + " | Size:
    }

    @Override
    public double getPrice() {
        return price;
    }
}

class GroceryProduct extends BaseProduct implements Product {
    String expiryDate;

    GroceryProduct(String name, double price, String expiryDate) {
        super(name, price);
        this.expiryDate = expiryDate;
    }

    @Override
    public void showDetails() {
        System.out.println("Grocery Product: " + name + " | Price: Rs. " + price + " | Expiry
    }

    @Override
    public double getPrice() {
        return price;
    }
}

public class ECommerceProductDemo {
    public static void main(String[] args) {
        Product p1 = new ElectronicProduct("Smartphone", 25000.0, 1);
        Product p2 = new ClothingProduct("T-Shirt", 599.0, "L");
        Product p3 = new GroceryProduct("Rice Bag", 450.0, "31-Dec-2026");

        p1.showDetails();
        p2.showDetails();
```

```java
class GroceryProduct extends BaseProduct implements Product {
    public void showDetails() {
    }

    @Override
    public double getPrice() {
        return price;
    }
}

public class ECommerceProductDemo {
    public static void main(String[] args) {
        Product p1 = new ElectronicProduct("Smartphone", 25000.0, 1);
        Product p2 = new ClothingProduct("T-Shirt", 599.0, "L");
        Product p3 = new GroceryProduct("Rice Bag", 450.0, "31-Dec-2026");

        p1.showDetails();
        p2.showDetails();
        p3.showDetails();

        double total = p1.getPrice() + p2.getPrice() + p3.getPrice();
        System.out.println("Total Cart Value: Rs. " + total);
    }
}
```

```
Electronic Product: Smartphone | Price: Rs. 25000.0 | Warranty: 1 years
Clothing Product: T-Shirt | Price: Rs. 599.0 | Size: L
Grocery Product: Rice Bag | Price: Rs. 450.0 | Expiry: 31-Dec-2026
Total Cart Value: Rs. 26049.0
```

Conclusion: Circle and Rectangle reuse common structure from Shape via inheritance, while Product interface lets unrelated product types (Electronic, Clothing, Grocery) share a common contract for polymorphic handling.

Exercise

1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.